

Falsifiable Predictions of the Evaporating Universe Model

Prediction	Observable	Current Status	Future Test
$w(z < 0.2) < -1$	BAO: DESI DR2	SUPPORTED (2.5σ)	DESI DR3
$H_0 = 73.2 \pm 2.5$	Cepheids/TRGB	SUPPORTED (0.06σ)	JWST distance ladder
$z_{trans} = 0.22 \pm 0.05$	$w(z)$ evolution	TESTABLE	DESI/Euclid
Soliton cores	Dwarf galaxy kinematics	SUPPORTED	JWST + VLT
$m_\phi \approx 5 \text{ MeV}$	Li-7 abundance	SUPPORTED	High-z quasars
ISW suppression	CMB-LSS cross-correlation	NEUTRAL	Planck + DESI
Void profiles	Void depth distribution	TESTABLE	DESI void catalog
σ_8 evolution	Weak lensing tomography	PARTIAL	Euclid + Rubin

Legend: SUPPORTED = consistent with data | TESTABLE = awaiting data | PARTIAL = partially consistent